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Discovery of Novel FXIa Inhibitory Peptides from Deep-Sea *Gigantidas platifrons* Hydrolysates: AI-Guided Screening, Anticoagulant Activity, and Mechanistic Insights

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ABSTRACT: The development of anticoagulants targeting coagulation factor XIa (FXIa) is a promising strategy to reduce thrombotic risk while minimizing bleeding complications. However, FXIa inhibitory peptides derived from the deep sea remain largely unexplored. In this study, pepsin hydrolysates from the deep-sea mussel *Gigantidas platifrons* exhibited anticoagulant effects through prolongation of activated partial thromboplastin time (APTT) and reducing erythrocyte and platelet aggregation in a zebrafish thrombosis model. Through bioactivity-guided fractionation combined with AI-assisted virtual screening, three novel FXIa inhibitory peptides were identified: VVTRVVR (IC_{50} = 523 μM), TVTRLSD (IC_{50} = 683 μM), and VSQPRSTL (IC_{50} = 695 μM). These peptides selectively inhibited FXIa without affecting FXa or thrombin and prolonged APTT. Structural analyses using AlphaFold3 modeling, molecular dynamics simulations, and quantum chemical calculations revealed that binding stability was driven by electrostatic interactions with Asp189 in the S1 pocket. This work provides a strategy for discovering anticoagulant peptides.

KEYWORDS: *anticoagulant activity, deep-sea peptides, coagulation factor XIa inhibitor, molecular dynamics simulations, gigantidas platifrons*

1. INTRODUCTION

Thrombotic diseases have become a critical concern in global public health due to their high incidence and mortality rates.[1] The coagulation system, comprising a series of tightly regulated enzymatic reactions predominantly mediated by serine proteases, plays a central role in the formation of venous thrombus.[2] The coagulation cascade consists of the intrinsic and extrinsic pathways, which converge at the activation of factor X (FX). Activated FX (FXa) further leads to thrombin-mediated conversion of soluble fibrinogen into insoluble fibrin and subsequent clot formation.[3] Disruption of this delicately balanced system can lead to either excessive bleeding or pathological thrombosis.[4] Currently available anticoagulants, including low molecular weight heparin (LMWH), apixaban, rivaroxaban, and warfarin, although effective, are often associated with undesirable side effects.[5] In recent years, increasing attention has been directed toward food-derived bioactive peptides as safer and more biocompatible alternatives for prevention and adjunctive treatment of cardiovascular diseases.[6] Generally, bioactive oligopeptides are short sequences derived from the nature or hydrolysis of food proteins, typically comprising 2 to 20 amino acids.[7] Based on their amino acid composition and sequence features, these peptides may exert regulatory effects on various biological processes by inhibiting key enzymes or interfering with essential cofactors.[8]

The ocean represents a valuable reservoir for discovery of novel bioactive peptides. Among the marine organisms, mollusks have shown great potential as sources of anticoagulant peptides.[9] Owing to their low-fat and high-protein composition, marine mollusks constitute a high-quality nutritional source well-suited for functional food applications.[10] They are rich in bioactive constituents such as polysaccharides, peptides, and proteins, which have been reported to exhibit diverse biological activities, including antioxidant, antitumor, and immunomodulatory effects.[11,12] Given their high nutritional value and functional properties, mollusks hold considerable potential for the development of novel food products.[13] Notably, several key anticoagulant peptides have been isolated and identified from shallow-water mollusks. Specifically, the peptides KNAENELGEVTVR, ELEDSLDSER, and EADIDGDGQVNYEEFVAMMTSK were isolated from *Mytilus edulis*.[14−16] The peptides TARNEANVNIY, NAESLRK, and IEELEELEAER were isolated from *Crassostrea gigas*.[5,17,18] Within this context, the potential of peptides from the deep-sea mollusks remains poorly understood. Mussels of the subfamily Bathymodiolinae are recognized as common and numerically dominant inhabitants of deep-sea cold seeps and hydrothermal vents,[19] where *Gigantidas platifrons* stands out as a representative and ecologically dominant species in the Western Pacific cold seep ecosystems.[20] Current research on *G. platifrons* has primarily focused on its genome and ecological adaptation, while little is known about its functional proteins and bioactive peptides.[21,22] This gap highlights its

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potential as a novel source of bioactive peptides with nutritional and health-promoting effects.

Factor Xa and thrombin are central mediators of hemostasis following vascular injury, yet anticoagulants targeting these factors often increase the risk of major bleeding in critical sites.[23] To balance efficacy with safety, attention has shifted to factor XIa (FXIa), a serine protease with a comparatively minor role in normal hemostasis but a significant contribution to thrombosis.[24,25] Targeting FXIa has been shown to preserve antithrombotic efficacy while minimizing bleeding risk, positioning it as an attractive therapeutic target.[26] Within the intrinsic pathway, FXIIa activates FXI to FXIa, which further converts FIX to FIXa; the resulting FIXa−FVIIIa complex then activates FX, therefore initiating thrombin generation.[27] Most FXIa inhibitors target its serine protease domain, which contains 238 residues and the canonical catalytic triad—His57, Asp102, and Ser195—essential for activity.[28] Multiple therapeutic modalities, including peptides, peptidomimetics and small-molecule inhibitors, have been explored to inhibit the FXI/FXIa pathway as potential alternatives for thrombosis treatment.[25] Existing natural peptide inhibitors are predominantly long-chain Kunitz-type peptides,[29] highlighting the promise of short-chain peptides as safe anticoagulants.

In this study, soluble extracts from the edible deep-sea mussel *G. platifrons* were subjected to pepsin hydrolysis, and the resulting peptides were evaluated for anticoagulant potential both *in vitro* and *in vivo*. The hydrolysates demonstrated significant inhibitory activity against FXIa, from which the most potent fraction was subsequently purified. Guided by AlphaFold3 (AF3)-assisted virtual screening, three peptides—TVTRLSD, VSQPRSTL and VVTRVVR—were identified as potential FXIa inhibitors. Their anticoagulant properties and selectivity were further validated through biochemical characterization. To gain mechanistic insight, molecular dynamics simulations combined with quantum chemical analyses were performed to reveal structural features underlying their inhibitory activities. Together, these findings highlight the deep-sea mussel *G. platifrons* as a valuable marine resource for the development of food-derived anticoagulant peptides and provide a mechanistic basis supporting their potential application in functional foods and nutraceuticals aimed at thrombosis prevention.

2. MATERIALS AND METHODS

2.1. Materials

The deep-sea mussel *G. platifrons* (GP) was collected in October 2017 from the cold seep site "F" (22°06′ N, 119°17′ E) in the South China Sea. The mussels were captured at a depth of 1100 m by the ROV *Faxian* aboard the scientific research vessel *Kexue*. Immediately after collection, these mussels were snap-frozen in liquid nitrogen for 24 h and subsequently stored at −80 °C until use. Prior to analysis, the samples were finely ground into powder.

2.2. Preparation of Pepsin-Digested GP Protein Hydrolysates (GPp)

GP powder was suspended in deionized water at a solid-to-liquid ratio of 1:10 (w/v). The resulting suspension was adjusted to pH 2 and incubated at 37 °C. Pepsin was added at an enzyme-to-substrate ratio of 1% (w/w), and hydrolysis was performed for 4 h. The reaction was terminated by heating at 95 °C for 10 min. Hydrolysates were centrifuged (8000*g*, 10 min), and the supernatants were freeze-dried and stored at −20 °C.

2.3. Fractionation of the GPp

GPp was first fractionated on a Sephadex G-15 column (2.6 cm × 100 cm). The sample (587.49 mg, 100 mg/mL) was eluted with deionized water at 0.7 mL/min and monitored at 220 and 280 nm. Fractions were collected according to the chromatographic profile and screened for FXIa inhibitory activity. After lyophilization, six fractions were obtained, with masses of 317.39, 104.71, 15.30, 9.62, 7.02, and 3.15 mg, respectively. Among them, G1 was the most active fraction and was selected for further purification.

The active fraction was subsequently purified by HPLC on a Sepax GP-C18 column (21.2 mm × 250 mm, 5 μm) using Milli-Q water (A) and acetonitrile (B) as mobile phases. The gradient was 10% B (0−10 min), increased to 90% B (10−25 min), and held at 90% B (25−35 min). The flow rate was 5 mL/min, detection wavelengths were 220 and 280 nm, and eluates were collected according to chromatographic peaks.

2.4. *In Silico* Screening of Anticoagulant Peptides

AF3 was used to predict FXIa−peptide complexes using the FXIa protease domain sequence from PDB ID 5E2O and each peptide sequence as inputs.[30] All predictions were performed using the default AlphaFold Server settings. No user-defined restraints, manually specified binding-site information, or additional prior interaction information was provided, and template/MSA settings were kept as the server defaults. For each peptide, the best-ranked model from AF3 was evaluated using ipTM, pLDDT, and binding pose. Peptides were retained when the complex ipTM was ≥ 0.8 and the average pLDDT of the peptide region was ≥ 70. The retained complexes were further evaluated using Chai-1 (https://lab.chaidiscovery.com/) and Protenix (https://protenix-server.com/) with default Web server settings. For each platform, the best-ranked model was selected, and the binding poses predicted by AF3, Chai-1, and Protenix were compared by structural visualization in PyMOL (v2.3). High-confidence peptides were selected for solid-phase synthesis by GenScript Biotech Co., Ltd. (Nanjing, China).

2.5. Evaluation of Anticoagulant Activity *In Vitro*

Inhibition of human FXIa, FXa, and α-thrombin by candidate peptides was evaluated using chromogenic substrates under physiological conditions (pH 7.4, 37 °C), based on a previously described approach with slight modifications.[31−33] The reaction buffer contained 50 mM Tris−HCl (pH 7.4), 150 mM NaCl, 0.1% PEG8000, and 0.02% Tween-80. For each assay, 170 μL buffer, 10 μL peptide solution, and 10 μL enzyme (FXIa: 0.765 nM; FXa: 1.09 nM; α-thrombin: 0.6 nM) were added to 96-well plates and incubated at 37 °C for 20 min. Reactions were initiated by adding 10 μL chromogenic substrate (S-2366, 345 μM; S-2765, 125 μM; S-2238, 50 μM) and allowed to proceed for 20 min. Absorbance at 405 nm was measured to determine residual enzymatic activity relative to peptide-free controls, and dose−response curves were fitted using a four-parameter logistic equation

$$Y = Y_0 + \frac{Y_M - Y_0}{1 + 10^{(\log[I]0 - \log IC_{50})(HS)}} \quad (1)$$

The parameters derived from the curve fitting included maximum (Y_M) and minimum (Y_0) activity values, the IC_{50}, and the Hill slope (HS), which describes the steepness of inhibition curves.

Clotting parameters were determined using a one-stage recalcification assay on a semiautomatic coagulation analyzer (HF6000−4, Hanfang, China).[34] Magnetic beads were added according to the manufacturer's instructions, and plasma and reagents were pre-equilibrated at 37 °C. For the PT assay, 10 μL of peptide was mixed with citrated goat plasma to 40 μL, incubated for 1 min at 37 °C, and coagulation was initiated by adding 60 μL of prewarmed PT reagent. For the TT assay, 10 μL of peptide was incubated with 45 μL of plasma for 1 min at 37 °C, followed by addition of 45 μL of TT reagent. For the APTT assay, 10 μL of peptide was mixed with 30 μL of plasma and 30 μL of APTT reagent (0.2% ellagic acid), incubated for 60 s at 37 °C, and clotting was triggered by adding 30 μL of 25 mM $CaCl_2$.

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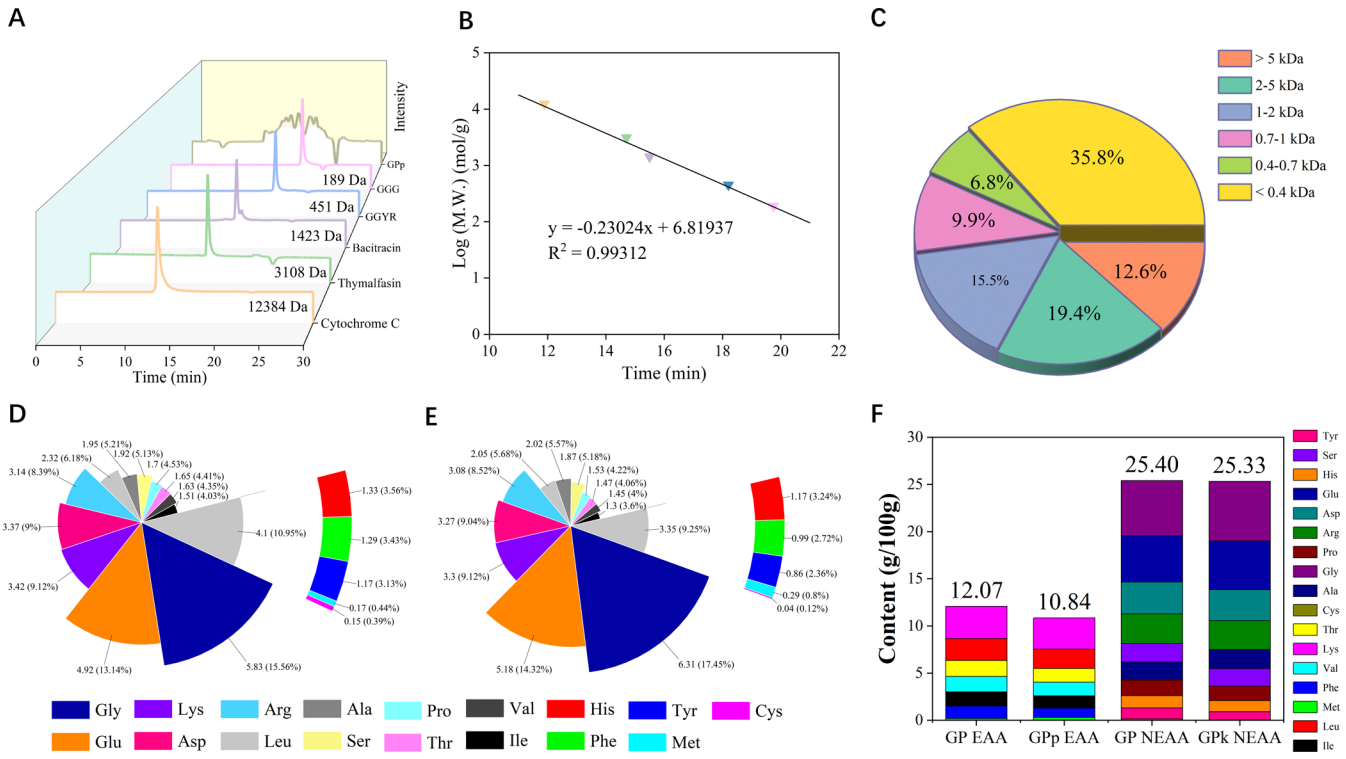


Figure 1. Molecular weight analysis and amino acid composition of the GPp. (A) The molecular weight distribution of the standards and GPp was determined using a TSK G2000 column. (B) Molecular weight curve fitting results for five standards. (C) Molecular weight distribution of GPp. Amino acid content of the (D) GP and (E) GPp. (F) Comparison of essential amino acids (EAA) and nonessential amino acids (NEAA) in GP and GPp. Trp was not determined because of its oxidation during amino acid composition analysis.

2.6. Evaluation of Antithrombotic Activity *In Vivo*

Tg(gata1:dSRed) and Tg(CD41:EGFP) zebrafish were maintained at 28.5 °C under standard conditions as previously described.[35] Fertilized embryos from natural spawning were maintained at 28.5 °C in recirculating system water supplemented with methylene blue. All animal procedures were approved by the Ethical and Animal Welfare Committee of the Institute of Oceanology, Chinese Academy of Sciences (Approval No. IOCAS20260114PPFA0004) and complied with institutional guidelines and the ARRIVE guidelines.

The zebrafish assay was conducted following Yang et al. with slight modifications.[36] At 2 dpf, Tg(gata1:dSRed) and Tg(CD41:EGFP) embryos were placed in 6-well plates (30 embryos per well) and pretreated with GPp (100, 200, or 400 μg/mL) for 24 h. Aspirin (15 μg/mL) and 0.1% DMSO served as positive and negative controls, respectively. Embryos were then exposed to 1.5 μM ARA for 80 min. Ten to 15 embryos per group were randomly selected for analysis. Erythrocyte accumulation in the heart and thrombocyte aggregation were imaged using a confocal microscope (LSM 900, Zeiss), and thrombus formation was quantified by integrated optical density (IOD) using ImageJ (v1.54p).

2.7. Quantum Chemical Calculations

Density functional theory (DFT) calculations were performed using ORCA 6.[37] Single-point energy calculation and electronic structure analyses were conducted at the B3LYP/def2-SVP level of theory.[38] Molecular electrostatic potential maps, HOMO−LUMO energies, and conceptual DFT descriptors were then obtained. Multiwfn 3.8[39] was used for data analysis, and molecular structures were visualized with VMD 1.9.4.[40]

2.8. Molecular Dynamics Simulation

Molecular dynamics (MD) simulations were performed using GROMACS 2024.2[41] to evaluate the stability and dynamic interactions of the FXIa−peptide complexes. Protein structures were obtained from AF3 models. For each peptide, the top three AF3-ranked complex structures were selected and subjected to three independent MD simulations, with one simulation performed for each selected structure. System topologies were generated using the CHARMM36 force field for FXIa and CGenFF for ligands.[42,43] Each complex was solvated in a dodecahedral TIP3P water box with a 10 Å buffer, neutralized with counterions, and energy-minimized (maximum force <1000 kJ/mol). After NVT and NPT equilibration at 310 K and 1 atm, 500 ns production simulations were conducted with snapshots saved every 10 ps. Binding free energies were calculated by MM/PBSA using gmx_MMPBSA[44] based on the last 100 ns, with per-residue decomposition to identify residues within 3 Å of the ligand.

2.9. Statistical Analysis

All experiments were performed at least in triplicate, and results are presented as mean ± SD. Data were analyzed using one-way ANOVA followed by multiple comparisons within GraphPad Prism 9.5. *p value <0.05 or **p value <0.01 was considered statistically significant.

3. RESULTS AND DISCUSSION

3.1. Physical and Chemical Properties Characterization of GPp

Direct enzymatic hydrolysis of *G. platifrons* protein using pepsin was performed to generate peptide-rich hydrolysates, yielding an extraction rate of 45.3%. Size distribution analysis was conducted using molecular weight standards ranging from 189 to 12 384 Da. The resulting calibration curve allowed characterization of peptide sizes in the hydrolysate (Figure 1A,B). The majority of peptides were below 1 kDa, with <0.4 kDa peptides representing the predominant fraction (Figure 1C). The prevalence of low-molecular-weight peptides indicates efficient generation of oligopeptides after pepsin digestion. Amino acid composition analysis further showed that glycine was the dominant residue in both untreated

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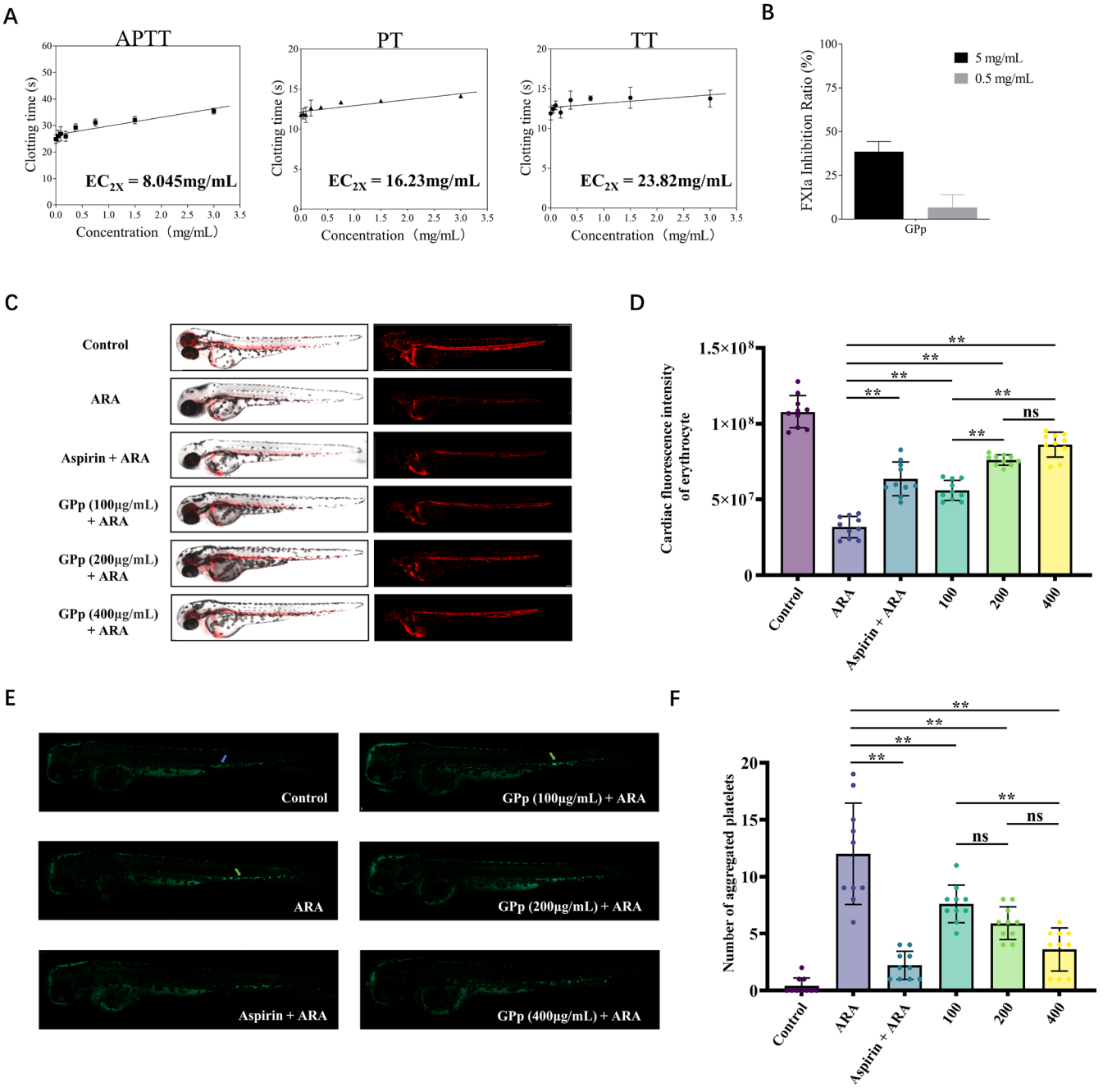


Figure 2. *In vitro* and *in vivo* anticoagulant effects of GPp. (A) Effects of different concentrations of GPp on the clotting times of APTT, PT, and TT, in goat plasma. APTT, activated partial thromboplastin time; PT, prothrombin time; and TT, thrombin time. (B) Inhibitory activity of GPp against FXIa at different concentrations. (C) Representative fluorescence images showing erythrocytes in 3 dpf Tg(gata1:dSRed) embryos. (D) Quantification of erythrocyte accumulation in the cardiac region. (E) Fluorescence visualization of circulating platelets in Tg(CD41:eGFP) embryos under control conditions, after ARA exposure, and following ARA exposure with GPp treatment. Green arrows highlight platelet aggregates, whereas blue arrows indicate circulating platelets. (F) Quantitative analysis of platelet aggregation. Data are presented as mean ± SD $^{**}p < 0.01$.

protein and pepsin hydrolysates (Figure 1D,E). This high glycine content may reflect compositional features associated with deep-sea adaptation, as amino acid composition and protein structural properties have been reported to contribute to protein function under high hydrostatic pressure.[45,46] The essential-to-nonessential amino acid (EAA/NEAA) ratio remained close to 45% (Figure 1F). Furthermore, the relatively high proportion of essential amino acids highlights the nutritional value of *G. platifrons* protein. Overall, the favorable compositional and nutritional profiles support the use of GPp low-molecular-weight peptides for subsequent anticoagulant screening.

3.2. *In Vitro* and *In Vivo* Coagulation Assays Revealing the Anticoagulant Effect of GPp

To evaluate the anticoagulant activity of GPp *in vitro*, APTT, PT and TT were measured. GPp led to significant prolongation of all three coagulation parameters (Figure 2A). Among them, APTT showed the greatest sensitivity, with an EC_{2X} value of 8.045 mg/mL. In addition, GPp exhibited a 40%

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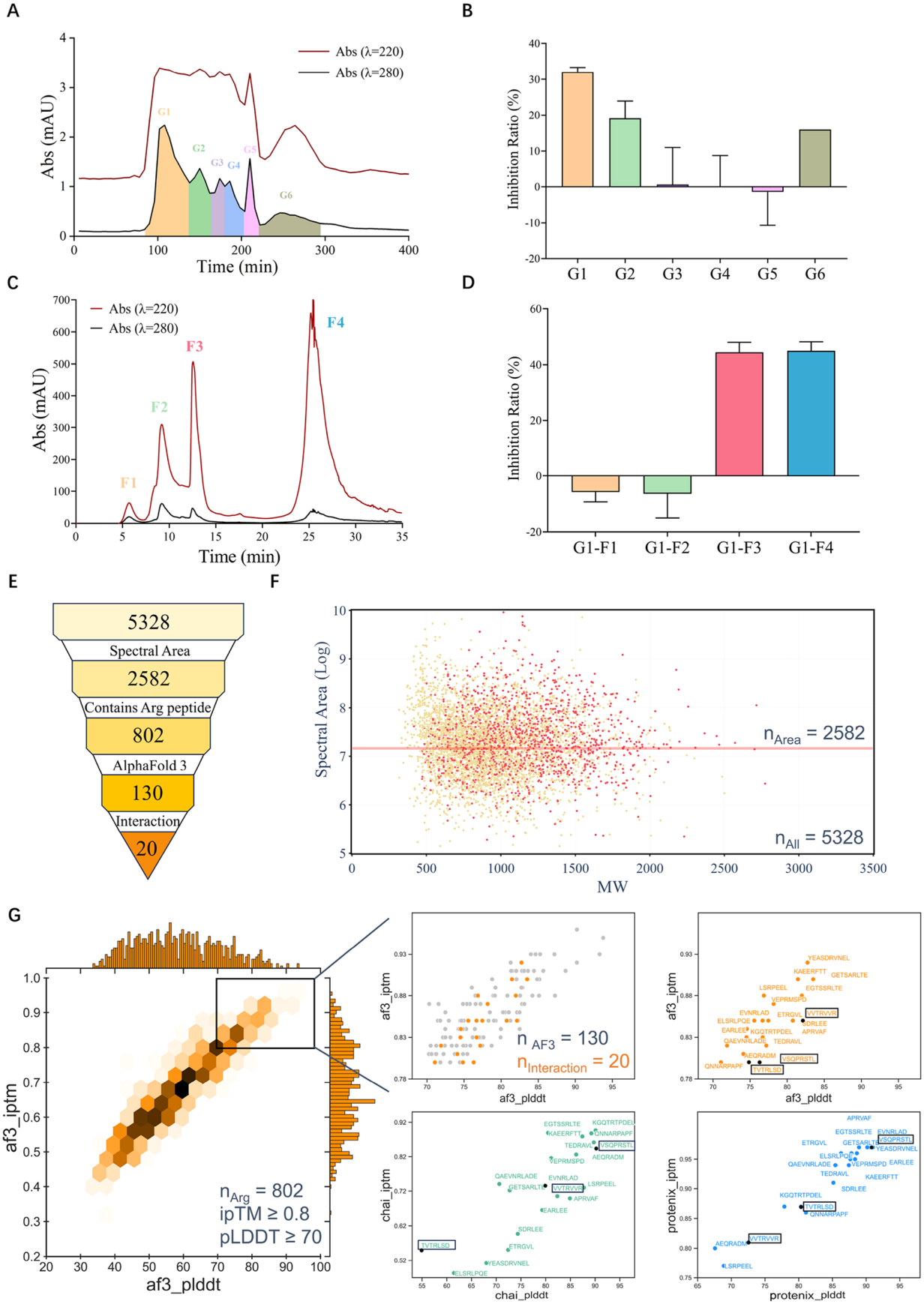


Figure 3. Bioactivity-guided fractionation and virtual screening of GPp. (A) Elution profiles of GPp monitored at 220 and 280 nm using Sephadex G-15 chromatography. (B) Inhibitory effects of G-15 fractions of GPp on FXIa activity at 3 mg/mL. (C) Elution profiles of the G1 fraction detected at 220 and 280 nm using a Sepax GP-C18 column. (D) FXIa inhibitory activity of subfractions derived from G1 at 3 mg/mL. (E)

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Figure 3. continued

Workflow of the AI-assisted virtual screening for peptide candidates. (F) The abundance and the molecular weights of the peptides identified by LC−MS/MS profiling. Red dots indicate peptides containing arginine residues. (G) Candidate peptide screening pipeline implemented with AF3, Chai-1 and Protenix. The black circle dots indicate validated FXIa inhibitory peptides.

Table 1. Properties of 20 Synthetic Peptides

no.	peptide sequence	*de novo* score	−10lg *P*	spectral area	MW (Da)	length
1	QAEVNRLADE	/	40.44	193250000	1143.6	10
2	GETSARLTE	/	29.13	10802000	962.5	9
3	YEASDRVNEL	/	27.09	48409000	1194.6	10
4	AEQRADM	/	25.27	228380000	819.4	7
5	KAEERFTT	/	22.55	46963000	980.5	8
6	SDRLEE	99	/	68008000	747.3	6
7	ETRGVL	99	/	243550000	673.4	6
8	LSRPEEL	97	/	51457000	842.4	7
9	VVTRVVR	97	/	74758000	827.5	7
10	TVTRLSD	95	/	23907000	790.4	7
11	VSQPRSTL	95	/	90035000	887.5	8
12	APRVAF	95	/	35283000	659.4	6
13	KGQTRTPDEL	95	/	7572300000	1144.6	10
14	EGTSSRLTE	93	/	388400000	978.5	9
15	EVNRLAD	92	/	34291000	815.4	7
16	QNNARPAPF	85	/	47182000	1015.5	9
17	TEDRAVL	83	/	129330000	802.4	7
18	EARLEE	81	/	26083000	745.4	6
19	ELSRLPQE	80	/	27760000	970.5	8
20	VEPRMSPD	80	/	48634000	815.5	8

inhibition rate against FXIa at a concentration of 5 mg/mL (Figure 2B). The pronounced effect of GPp on APTT suggests modulation of the intrinsic coagulation pathway, consistent with previous findings in pepsin hydrolysates of *Crassostrea gigas*.[17] The observed FXIa inhibition further supports this mechanism. Based on these results, fractionation and purification strategies are warranted to isolate specific anticoagulant peptides from GPp.

Arachidonic acid (ARA) serves as a potent platelet aggregation inducer through the stimulation of endogenous ADP release and subsequent platelet activation, making it a widely utilized thrombogenic agent in experimental animal models of thrombosis.[47] Conversely, aspirin functions as an antiplatelet agent that suppresses platelet activation and reduces aggregation.[48] To investigate the antithrombotic properties of GPp, we used an ARA-induced zebrafish thrombosis model.

The validity of the ARA-induced thrombosis model was confirmed by assessing erythrocyte aggregation and cardiac fluorescence intensity in Tg(gata1:dSRed) zebrafish embryos. ARA treatment induced obvious thrombotic phenotypes, including reduced erythrocyte accumulation in the cardiac region and increased platelet aggregation, confirming successful thrombosis induction. GPp pretreatment restored blood flow in a dose-dependent manner (Figure 2C). Correspondingly, ARA exposure significantly decreased erythrocyte levels in the cardiac chambers (Figure 2D), whereas pretreatment with 200 μg/mL of GPp markedly restored cardiac erythrocyte counts, showing stronger efficacy than 15 μg/mL of aspirin. To further investigate the antithrombotic effects of GPp, platelet behavior was examined in Tg (CD41:eGFP) zebrafish. ARA treatment induced pronounced peripheral platelet aggregation (Figure 2E). GPp pretreatment effectively inhibited this response in a dose-dependent manner, as confirmed by fluorescence quantification showing significantly reduced platelet aggregation numbers (Figure 2F). Our results demonstrate that GPp exerts potent antithrombotic activity, notably by restoring cardiac blood flow and suppressing pathological platelet aggregation in a concentration-dependent manner, underscoring its potential as a protective agent against vascular thrombotic complications.

3.3. Identification of FXIa-Inhibitory Peptides via Fractionation and AI-Based Virtual Screening

A bioactivity-guided fractionation strategy was used to isolate FXIa-inhibitory peptides from GPp. Gel filtration chromatography separated GPp into six fractions—G1−G6 (Figure 3A), among which G1 showed the strongest FXIa inhibition (32.03 ± 1.05% at 3 mg/mL) (Figure 3B), indicating enrichment of active components. Subsequent purification of G1 by RP-HPLC yielded four subfractions (F1−F4) (Figure 3C), with F3 and F4 exhibiting the highest inhibitory activity (Figure 3D). LC−MS/MS analysis of F3 identified 5328 peptide sequences for subsequent virtual screening. Peptide identification was filtered using a *de novo* score >80 for *de novo*−sequenced peptides and a −10lg *P* value >20 for database-matched peptides. The screening workflow is shown in Figure 3E. Based on relative abundance, the top 2582 peptides were selected (Figure 3F), of which 802 arginine-containing peptides were retained for further analysis.[49]

Accurate modeling of protein−peptide interactions has historically posed significant challenges, as conventional computational approaches often fail to capture the structural complexity of these systems. Recent advances in deep learning−based prediction frameworks have begun to overcome these limitations.[50] AlphaFold2, for instance, substantially improved structural accuracy for protein−peptide

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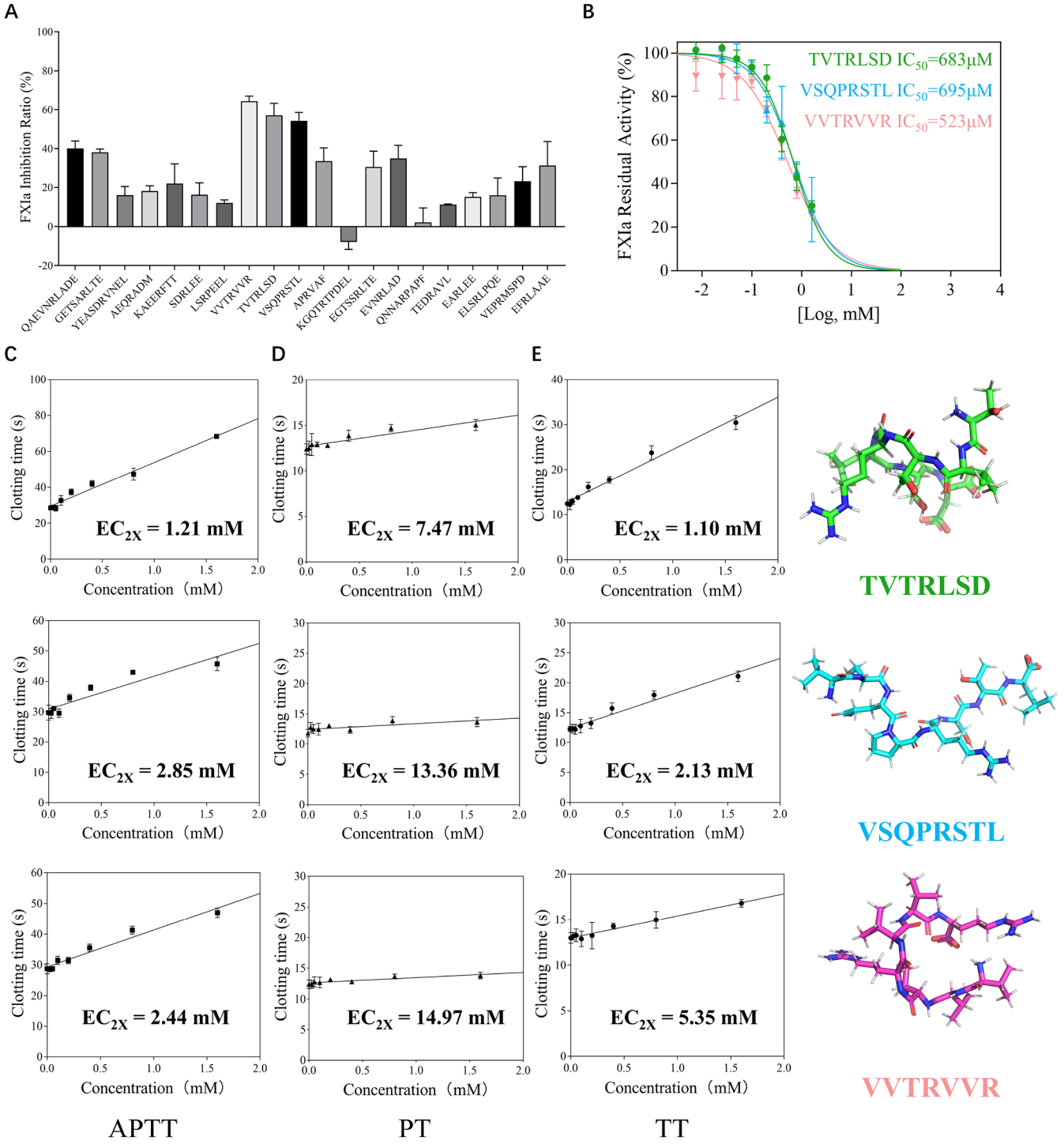


Figure 4. Anticoagulant activity of candidate peptides. (A) FXIa inhibitory activities of the 20 peptides at the concentrations of 0.8 mM. (B) Dose-dependent curves of TVTRLSD, VSQPRSTL and VVTRVVR on FXIa. IC_{50} values were calculated based on the fitted curves. Effects of varying concentrations of the three peptides on (C) APTT, (D) PT and (E) TT in goat plasma.

complexes, while its successor AF3 has shown strong potential for modeling protein−peptide complexes.[30] Parallel developments have also yielded alternative implementations such as Chai-1[51] and Protenix,[52] which provide open-source code and data sets for broader application and methodological refinement. Moreover, integrative strategies that combine multiple next-generation predictors and incorporate diverse structural confidence metrics—including Predicted Local Distance Difference Test (pLDDT), interface predicted TM-score (ipTM), and Root Mean Square Deviation (RMSD)—have been shown to further enhance the identification of reliable complex models.[53] Nevertheless, despite the paradigm-shifting advances of AF3 in protein−ligand complex prediction and its strong potential for virtual screening, most existing studies remain largely focused on pose reconstruction of known complexes, with systematic exploration and end-to-end application in peptide discovery from complex biological samples still being limited.[54] In the present study, AF3 was

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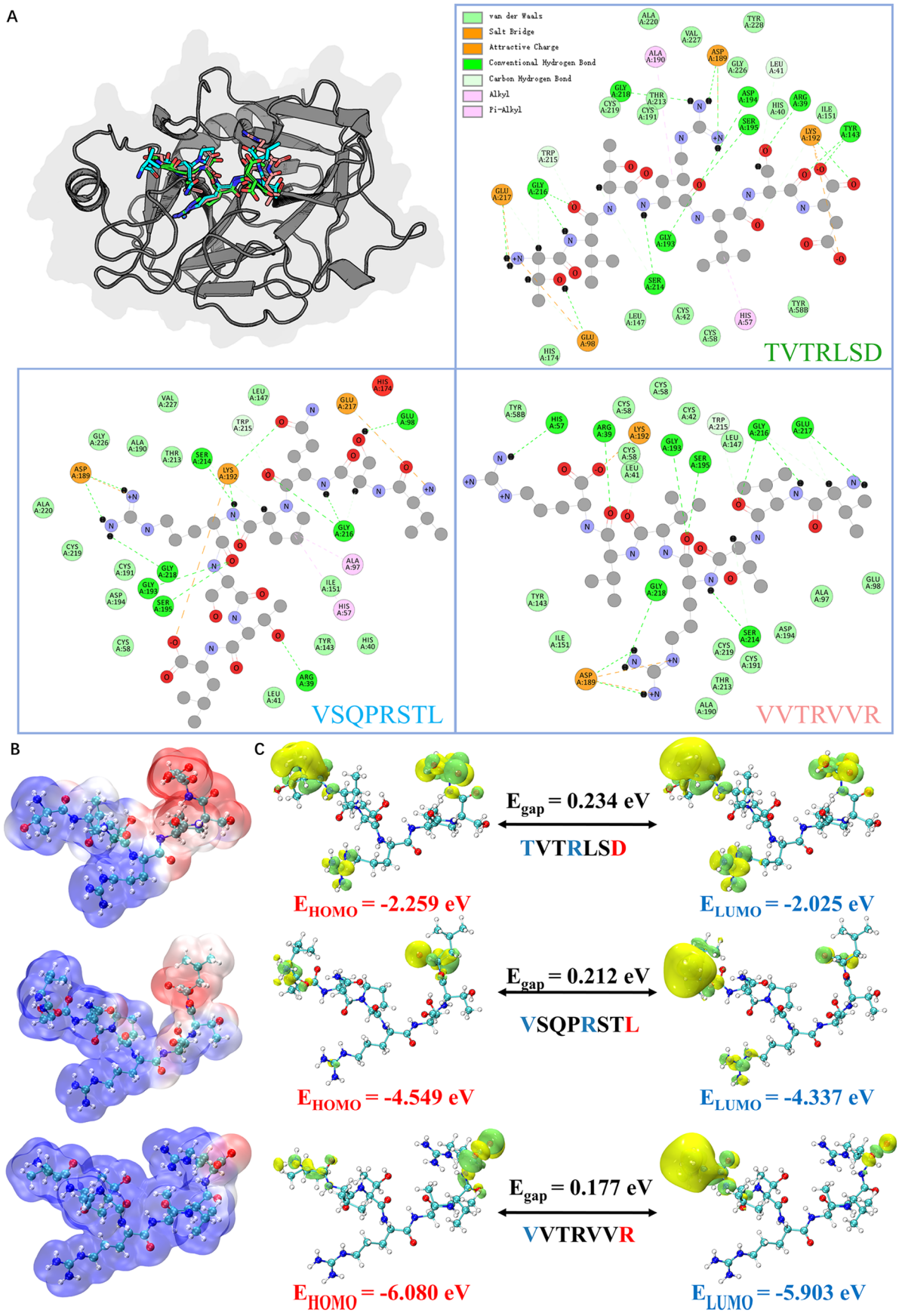


Figure 5. Molecular docking and quantum chemical characterization of FXIa−peptide complexes. (A) Structural modeling and interaction analysis of three FXIa−peptide complexes. Note: Green represents TVTRLSD; blue represents VSQPRSTL; bright red represents VVTRVVR. (B)

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Figure 5. continued

Electrostatic potential maps of the three peptides. (C) Highest occupied molecular orbital (HOMO) and lowest unoccupied molecular orbital (LUMO) surface distributions of the three peptides.

applied to predict interactions between 802 candidate peptides and FXIa (Figure 3G). Peptides meeting the criteria of ipTM $\geq$ 0.8 and pLDDT $\geq$ 70 were retained, and these candidates also achieved high confidence scores when validated with Chai-1 and Protenix. Cross-platform visualization and interaction analysis resulted in the identification of 20 peptides with potential anticoagulant activity (Table 1). These peptides, ranging from 5 to 10 residues in length, were among the more abundant components in the data set. Most of them occupied the S1, S1′, and S2′ subsites of FXIa, resembling the binding modes of small-molecule inhibitors.[55] Their pharmacokinetic and safety properties were assessed using ADMETlab 3.0,[56] and the radar plot evaluation indicated that all candidates fell within acceptable physicochemical parameters (Figure S1). These results suggest that the selected peptide candidates possess acceptable preliminary physicochemical properties for further evaluation.

3.4. *In Vitro* Validation of Anticoagulant Activity of Candidate Peptides

A panel of 20 peptides was successfully synthesized with purities exceeding 95%. Their inhibitory activities against FXIa, FXa, and α-thrombin were assessed using three chromogenic substrates. Among them, VVTRVVR, TVTRLSD, and VSQPRSTL showed pronounced inhibition of FXIa, each reducing enzymatic activity by more than 50% at 0.8 mM, while exerting negligible effects on FXa and α-thrombin (Figures 4A and S2). The concentration−response curves of these peptides conformed to a logistic model, yielding IC_{50} values of 523, 683, and 695 μM, respectively (Figure 4B). All three peptides displayed marked anticoagulant effects *in vitro*. TVTRLSD showed the strongest activity, prolonging APTT at 1.12 mM, while VSQPRSTL and VVTRVVR required 2.85 and 2.44 mM, respectively (Figure 4C). In contrast, their effects on PT and TT occurred only at much higher concentrations (Figure 4D,E), indicating a preferential action on the intrinsic coagulation pathway.

The strong selectivity observed for VVTRVVR, TVTRLSD, and VSQPRSTL underscores their potential as promising leads for the development of safer peptide-based anticoagulants. Notably, the IC_{50} values of VVTRVVR and TVTRLSD were lower than that of PRNIF (IC_{50} = 670 μM), a pentapeptide previously reported as an FXIa inhibitor.[57] Importantly, three candidates of the 20 synthesized peptides exhibited more than 50% inhibition of FXIa at 0.8 mM, corresponding to a hit rate of about 15%. This performance represents a clear improvement over our previous virtual screening efforts based on physics-based docking using smina (7.7% hit rate)[57] and further supports the notion that AF3 enables a substantially higher enrichment of functional peptide inhibitors. Two peptides, a decapeptide (ELEDSLDSER) and a tridecapeptide (KNAENELGEVTVR), previously isolated from shallow-water mussel hydrolysates, have been reported as effective thrombin inhibitors.[14,16] However, to date, no FXIa-inhibitory oligopeptides have been identified from marine organisms, highlighting the novelty of VVTRVVR, TVTRLSD, and VSQPRSTL as a marine-derived FXIa inhibitors.

To preliminarily address the physiological stability of the three validated peptides, *in silico* gastrointestinal digestion and plasma half-life prediction were further performed (Table S1). BIOPEP-UWM analysis suggested that TVTRLSD could be cleaved by both pepsin and trypsin, whereas VSQPRSTL and VVTRVVR were predicted to resist pepsin digestion but could be cleaved by trypsin. PlifePred analysis predicted plasma half-lives of 767.01, 932.31, and 834.81s for TVTRLSD, VSQPRSTL, and VVTRVVR, respectively. These results indicate that the three peptides may exhibit limited proteolytic stability under physiological conditions.

3.5. Computational Binding Analysis

Structural prediction with AF3 revealed that all three FXIa-inhibitory peptides occupied the canonical binding clefts of FXIa, spanning the S1, S1′ and S2′ pockets (Figure 5A). In all cases, the central arginine residue was positioned identically within the S1 pocket, forming a salt bridge with Asp189, thereby anchoring the peptide to the catalytic site. The N-terminus faces the edge of the S1 pocket, with its terminal amino group forming salt bridge interactions with Glu217, while the C-terminus consistently extends into the S2′ pocket. Despite this conserved binding mode, structural deviations were observed: the peptide backbones aligned closely in the S1 regions, likely due to the stable salt bridge formed by Asp189 and Glu217. However, significant divergence occurs in the S1′ and S2′ pockets, indicating the presence of differential stabilization interactions. Hydrogen bond analysis further highlighted both shared and distinct contact profiles. TVTRLSD formed hydrogen bonds with Asp189, Gly193, Ser195, Gly216, Gly218, Ser214, and Glu217; VSQPRSTL with Glu98, Gly193, Ser195, Lys192, Ser214, Gly216, and Gly218; and VVTRVVR with Arg39, Gly193, Ser195, His57, Asp194, Ser214, Gly216, Gly218, and Glu217. These hydrogen bonds were primarily concentrated within the S1, S1′, and S2′ regions.

This conserved anchoring of arginine in the S1 pocket mirrors previously reported FXIa inhibitors,[58] while the additional salt bridge between the peptide N-terminal amino and Glu217—absent in earlier peptides such as PRNIF—likely contributes to the enhanced inhibitory potency observed for these novel peptides.[57] The superior activity of VVTRVVR likely arises from its ability to form additional salt bridges with Glu98, and Arg39, together with the largest number of hydrogen bond and salt bridge interactions, resulting in a more stable binding interface with FXIa.

Molecular docking showed that all three peptides formed stable ionic interactions with FXIa through three key anchoring points: the N-terminal residue interacting with Glu217, the arginine guanidinium group binding Asp189 in the S1 pocket, and the C-terminal carboxyl group engaging basic residues in the S2′ pocket. These interactions help stabilize the peptides within the active site and reduce dissociation. Electrostatic potential mapping revealed that the peptides are predominantly positively charged (Figure 5B), particularly around arginine residues in the S1 pocket, complementing the negatively charged catalytic region of FXIa.[59] Meanwhile, the negatively charged C-terminus forms salt bridges with Arg39

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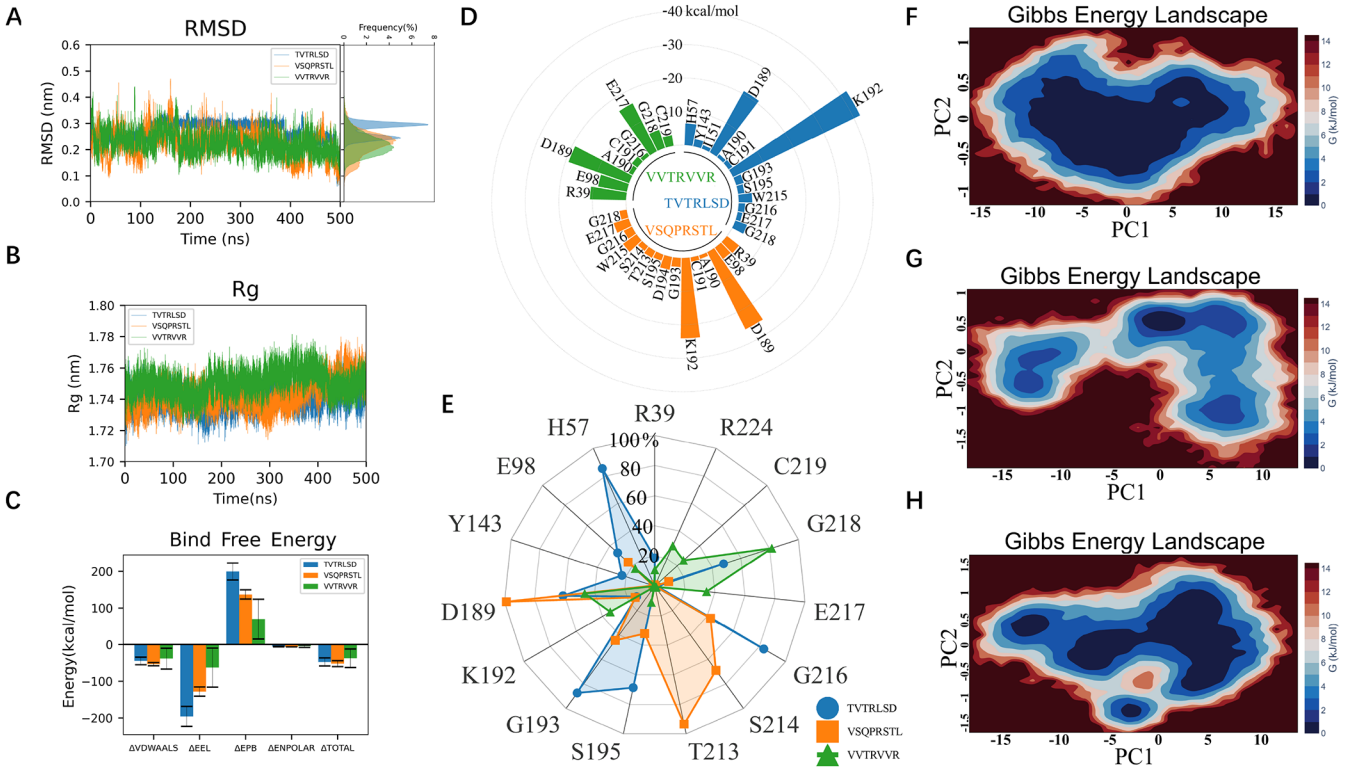


Figure 6. Molecular dynamics simulations of peptide−FXIa complexes over 500 ns. (A) Root mean square deviation (RMSD) and (B) radius of gyration (R_g) analyses of the complexes. (C) The total binding free energy and (D) per-residue binding free energy decomposition of the three peptides within the final 100 ns. (E) Hydrogen bond occupancy throughout the 500 ns trajectory. Free energy landscapes (FEL) associated with the binding of (F) TVTRLSD, (G) VSQPRSTL, and (H) VVTRVVR to FXIa.

and Lys192 in the S2′ pocket, further stabilizing binding. Consistently, quantum chemical calculations showed relatively small HOMO−LUMO energy gaps (Figure 5C), indicating high electronic reactivity and charge-transfer potential. The HOMO localized on the N-terminal amino and guanidinium groups, while the LUMO was mainly distributed on the C-terminal carboxyl group, supporting the N- to C-terminal interaction pattern observed in docking. Together, stable ionic interactions, electrostatic complementarity, and favorable electronic properties likely contribute to the strong and selective inhibition of FXIa by these peptides.

3.6. Molecular Dynamics Simulations Analysis

To further evaluate the dynamic behavior and local stability of the predicted FXIa−peptide complexes, 500 ns molecular dynamics simulations were performed. Throughout the trajectories, the peptide backbones remained stable (Figure 6A). Among the three peptides, TVTRLSD showed the highest RMSD value of 2.73 Å, whereas VSQPRSTL and VVTRVVR exhibited lower and very similar RMSD values of 2.24 and 2.23 Å, respectively, suggesting relatively smaller overall structural deviations during the simulation. Notably, TVTRLSD displayed two long-lived conformational states, indicating more evident conformational rearrangement during the MD trajectory. The radius of gyration (R_g) of all three proteins remained nearly constant (Figure 6B), and the RMSF profiles of the active-site residues were relatively low (Figure S3A), indicating limited structural fluctuations around the binding region. In addition, both the solvent-accessible surface area (SASA) (Figure S3B) and the center-of-mass distance between FXIa and the peptides (Figure S3C) remained relatively stable, supporting the persistence of the predicted peptide-binding poses under the simulation conditions. Dynamic cross-correlation analysis (DCCM) further revealed altered correlated motions among FXIa residues upon peptide binding (Figure S3D). Collectively, these findings suggest that the AF3-predicted FXIa−peptide complexes maintained local stability during the simulations. However, because short peptides are flexible and may adopt multiple binding conformations, these MD results should be interpreted as supporting the local stability and plausibility of the predicted FXIa−peptide binding poses, rather than as definitive structural validation of the binding modes.

Molecular mechanics/Poisson−Boltzmann surface area (MM/PBSA) analysis was employed to estimate the binding free energies of the FXIa−peptide complexes,[60] using the last 100 ns of the simulation trajectories. For each peptide, the binding free energy was calculated from the three independent MD simulations initiated from the top three AF3-ranked complex structures. The calculated binding free energies for TVTRLSD, VSQPRSTL, and VVTRVVR were −47.81 ± 10.77, −52.46 ± 8.09, and −37.76 ± 25.43 kcal/mol, respectively (Figure 6C), suggesting favorable estimated binding energies. These values were comparable to or lower than those reported for previously identified FXIa-inhibitory peptides PRNIF (−38.26 kcal/mol) and GNDRCL (−34.00 kcal/mol),[57] suggesting potential binding affinity of the present peptides. Nevertheless, because MM/PBSA is an end point approximation and the peptide ligands are flexible, these values should be interpreted cautiously as qualitative energetic support rather than definitive affinity rankings. Electrostatic contributions were found to play a major role in stabilizing the complexes. Residue decomposition analysis highlighted several

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charged residues—Asp189, Lys192 and Glu217—as key contributors to peptide binding (Figure 6D), with Asp189 consistently exerting a strong stabilizing effect across all three peptides. Notably, although VVTRVVR showed weaker interaction with Lys192, its favorable interactions with Glu98, Glu217 and Asp189 within the S1 pocket indicate particularly stable anchoring at this site. This observation aligns with the predominantly positive electrostatic potential of VVTRVVR and the localization of its HOMO on the N-terminal region, which enhances its ability to engage in electrostatic interactions with negatively charged residues of FXIa.

Analysis of hydrogen bond occupancy during the MD simulations revealed that the three FXIa-inhibitory peptides displayed distinct interaction patterns, reflecting differences in their binding modes (Figure S3E). For TVTRLSD, hydrogen bonds with His57, Asp189, Gly193, Ser195, and Gly216 were highly persistent, while VSQPRSTL preferentially interacted with Asp189, Thr213 and Ser214. In contrast, VVTRVVR exhibited a simpler pattern, maintaining stable hydrogen bonds only with Asp189 and Gly218 (Figure 6E). Despite these differences, the majority of interacting residues were concentrated in the S1 pocket, indicating that this region serves as the principal anchoring site for peptide binding. This observation is consistent with the docking results showing a conserved orientation of the three peptides within the S1 pocket, but weaker stabilization in the S1′ and S2′ pockets. Such findings suggest that future structural modifications could focus on enhancing interactions within S1′ and S2′ to improve binding stability. Notably, Asp189 was the only residue forming high-occupancy hydrogen bonds with all three peptides, reinforcing its essential role in mediating peptide recognition. This result aligns well with earlier studies on peptidomimetic FXIa inhibitors, which also identified Asp189 in the S1 pocket as a critical determinant of inhibition.[58] Together with the binding free energy analysis, these findings highlight Asp189 as a conserved hotspot for peptide-based inhibition of FXIa, while differences in auxiliary hydrogen bonding networks across the peptides may partly account for their varying inhibitory potency.

Principal Component Analysis (PCA) was applied to reduce the dimensionality of atomic motions and emphasize the dominant collective conformational changes within the system.[61] PCA was further employed to characterize the conformational dynamics of the peptides during MD simulations, and a three-dimensional free energy landscape (FEL) was constructed based on the first two principal components, PC1 and PC2 (Figures 6F–H and S3F). The FEL analysis showed that the three peptides sampled multiple metastable conformational states during the simulations, consistent with the intrinsic flexibility of short peptides. TVTRLSD displayed two relatively long-lived conformational states, in agreement with its larger RMSD fluctuation and conformational rearrangement. VSQPRSTL and VVTRVVR also occupied several local energy minima, suggesting conformational heterogeneity while maintaining the overall predicted binding poses. Collectively, these results indicate that the three peptides can maintain locally stable FXIa-bound states under the simulation conditions, while also exhibiting peptide-specific conformational flexibility.

In summary, the three *de novo* identified peptides bind stably to FXIa mainly through anchoring interactions within the S1 pocket, with Asp189 serving as a conserved recognition hotspot. VSQPRSTL showed the strongest binding affinity, consistent with its low binding free energy and a single energy basin in the FEL, indicating a stable conformational state. In contrast, VSQPRSTL and VVTRVVR exhibited greater conformational heterogeneity, reflected by multiple local minima and more dynamic hydrogen-bonding networks, which may contribute to their weaker inhibitory potential. Notably, VVTRVVR partly compensates for fewer hydrogen bonds through favorable electrostatic interactions with Glu98, Glu217, and Asp189, emphasizing the role of complementary charge interactions in complex stabilization. However, the limited engagement of the S1′ and S2′ pockets suggests that these peptides are not yet fully optimized, leaving room for sequence engineering to enhance interactions with these auxiliary pockets and improve inhibitory potency.

These findings complement the experimental identification of three novel FXIa inhibitory peptides—TVTRLSD, VSQPRSTL, and VVTRVVR—from deep-sea *G. platifrons* hydrolysates through an integrated workflow combining enzymatic digestion, bioactivity-guided purification, AI-assisted screening, and experimental validation. Importantly, the hydrolysate also exhibited significant antithrombotic activity in a zebrafish thrombosis model, providing *in vivo* support for the anticoagulant potential of these marine-derived peptides. Together with the mechanistic insights obtained from molecular simulations and quantum chemical analyses, this study highlights the value of AI-assisted peptide discovery platforms, such as AF3 and related tools, for accelerating the identification of functional anticoagulant peptides from complex biological resources.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.jafc.6c04005.

The information on chemical reagents used in this study, along with experimental methods for molecular weight determination, amino acid composition analysis, and LC–MS/MS identification, as well as supplementary figures referenced in the main text and computational input files (PDF)

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Notes

The authors declare no competing financial interest.

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